Distribution Narrowing Under Synthetic Data Contamination: A Multi-Seed Study on Linguistic Novelty Comprehension

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Abstract

As synthetically generated text proliferates online, each successive generation of language models risks training on the outputs of its predecessors. Prior work has shown that this recursive loop degrades *generation* quality; we ask whether it also degrades *comprehension*—the ability to process genuinely novel linguistic constructions the model has never seen. We fine-tune GPT-2 (124M) on WikiText-103 across three generations with synthetic contamination ratios from 0% to 100%, measuring the *log-probability gap* between standard and novel text as a continuous proxy for distributional competence (n=3 seeds, all results mean $\pm$ std). Under shallow fine-tuning (500 steps), moderate contamination (10–50%) produces distribution *narrowing*: the model becomes more confident on standard text while novel-text processing is preserved. Crucially, generation diversity collapses even at 10% contamination (Distinct-1 drops 37%), revealing an asymmetry where generation degrades before comprehension. Under deep fine-tuning (5,000 steps), this buffer breaks: at 100% contamination the gap widens to 2.30 ± 0.09 ($p < 0.001$), driven by a sharp decline in novel-text log-probability ($-7.16 \rightarrow -8.09$ nats), with perplexity collapsing to $1{,}482 \pm 30$ by generation 2. We further identify a degenerate collapse threshold (perplexity $> 1{,}400$) beyond which log-probability gap metrics become unreliable. Our results establish that pretrained representations are a buffer, not a firewall: they protect novel-text comprehension under light training but yield under sufficient optimization pressure on contaminated data. We recommend Distinct-N monitoring as a practical early-warning diagnostic for synthetic contamination in training pipelines.

1 Introduction

Over 50% of internet text may be machine-generated by 2026 [Hataya et al., 2023], and each new generation of language models [Vaswani et al., 2017, Radford et al., 2019, Brown et al., 2020] risks training on the outputs of its predecessors. Shumailov et al. [2024b] demonstrated that this recursive training loop causes *model collapse*: the tails of the original data distribution vanish irreversibly, and output quality degrades to repetitive, incoherent text. Alemohammad et al. [2023] formalized a complementary result—self-consuming generative models undergo "Model Autophagy Disorder," progressively losing distributional diversity regardless of architecture.

This paper asks a stronger question: does synthetic contamination destroy a model's ability to *comprehend* genuinely novel linguistic constructions—not just degrade its outputs, but eliminate the representational capacity for out-of-distribution language? If so, the consequence is not merely worse generation quality but a structural competence ceiling that constrains what future models can understand.

We test this hypothesis by fine-tuning GPT-2 [Radford et al., 2019] on WikiText-103 [Merity et al., 2017] with controlled synthetic contamination across multiple generations, replicated with n=3 random seeds. Our contributions are:

1. We introduce the *log-probability gap*, a continuous metric that measures the differential between a model's processing of standard versus novel text, and its *critical-span* variant that isolates the novel construction tokens from surrounding context.
2. We demonstrate dose-dependent gap widening under synthetic contamination with statistical testing (n=3 seeds), establishing that the effect is robust to random variation.
3. We decompose the gap into its constituent parts, revealing two distinct mechanisms: distribution *narrowing* at moderate contamination (10–50%) driven by improved standard-text confidence, and genuine competence *loss* at 100% driven by degraded novel-text processing.
4. We show that deeper fine-tuning (5,000 steps) breaks through the pretrained buffer that protects novel-text comprehension under shallow training (500 steps), establishing that pretrained representations are a buffer, not a firewall.
5. We identify a degenerate collapse threshold (perplexity $> 1{,}400$) beyond which log-probability gap metrics become unreliable, and we characterize the mechanism by which distributional collapse compresses the gap.

The remainder of this paper is organized as follows. Section 2 surveys prior work on model collapse and synthetic data quality. Section 3 describes our experimental setup, metrics, and benchmark. Section 4 presents results under shallow fine-tuning, and Section 5 extends to deep fine-tuning where the pretrained buffer breaks. Section 6 synthesizes the findings and discusses limitations, and Section 7 concludes.

2 Related Work

Model collapse. Shumailov et al. [2024b] established the foundational result: generative models trained recursively on their own outputs lose the tails of the original distribution. They demonstrated this across VAEs, GMMs, and LLMs (OPT-125M), showing that model collapse is architecture-general. Their protocol used full data replacement at each generation—the maximally pessimistic scenario. A companion paper [Shumailov et al., 2024a] further analyzed the conditions under which collapse is inevitable in recursive self-training, connecting the phenomenon to well-studied distribution shift dynamics [Quiñonero-Candela et al., 2009, Moreno-Torres et al., 2012].

Self-consuming generative models. Alemohammad et al. [2023] provided a complementary theoretical analysis, showing that self-consuming generative models "go MAD" (Model Autophagy Disorder), with distribution tails progressively lost across generations regardless of the generative model family. Bohacek and Braun [2023] demonstrated analogous collapse in image generation, coining the term "nepotistic training" for models trained on predecessor outputs. Dohmatob et al. [2024b] formalized conditions for *strong* model collapse where even the mean of the distribution shifts, not only the tails.

Synthetic data quality and contamination. Hataya et al. [2023] asked whether large-scale generative models will corrupt future datasets, finding that even small fractions of synthetic data in training corpora can shift learned distributions. Martínez et al. [2023] demonstrated distribution shift under synthetic augmentation in classification tasks, and Briesch et al. [2023] showed progressive quality degradation in synthetic data pipelines. Gudibande et al. [2023] warned that models trained to imitate proprietary LLM outputs learn surface-level stylistic patterns without acquiring the underlying capabilities—a form of distribution narrowing consistent with our findings. Long et al. [2024] found that neural retrievers are biased toward LLM-generated text, suggesting that contamination effects propagate beyond training into downstream information access.

On the constructive side, Liu et al. [2024] surveyed best practices for synthetic data in language model training, identifying data quality filtering as critical for avoiding degradation. Longpre et al. [2023] measured how data age, domain coverage, and quality affect pretraining outcomes, and Penedo et al. [2023] demonstrated that careful curation of web data can match curated corpora—suggesting that contamination effects depend on the quality pipeline, not merely the synthetic fraction.

Mitigating collapse through data mixing. Gerstgrasser et al. [2024] showed that if real and synthetic data *accumulate* rather than replace each other, collapse can be avoided. Dohmatob et al. [2024a] provided statistical formalization: collapse is unavoidable with 100% synthetic training but can be bounded below a threshold when mixing real and synthetic data. Our experiment uses mixed contamination ratios (0–100%) to map the full degradation curve rather than studying only the extremes, directly testing the predictions of these theoretical frameworks.

Fine-tuning and catastrophic forgetting. Our finding that pretrained representations buffer against contamination under shallow training but yield under deep training connects to the broader literature on fine-tuning dynamics. Kirkpatrick et al. [2017] identified catastrophic forgetting as a general risk when neural networks are trained sequentially on different distributions. Aghajanyan et al. [2021] showed that fine-tuning operates in a low-dimensional subspace of the full parameter space, suggesting that shallow fine-tuning may preserve pretrained representations precisely because it lacks the degrees of freedom to overwrite them. Kumar et al. [2022] demonstrated that standard fine-tuning can *distort* pretrained features and degrade out-of-distribution performance—a mechanism consistent with our deep-training results. Lee et al. [2022] proposed surgical fine-tuning as a mitigation, selectively updating only the layers most relevant to the target task.

Evaluation of language model quality. Perplexity has been a standard evaluation metric for language models since Jelinek et al. [1977]. For generation quality, Li et al. [2016] introduced Distinct-N as a diversity measure, and Holtzman et al. [2020] characterized the degenerate repetition patterns that arise from standard decoding, motivating the nucleus sampling strategy [Fan et al., 2018] we use for synthetic data generation. Pillutla et al. [2021] proposed MAUVE scores for measuring distributional gaps between neural and human text; our log-probability gap metric is conceptually related but operationalized differently—we measure the model's own differential processing of standard versus novel held-out text, rather than comparing generated text to a reference corpus.

Comprehension versus generation. Prior work has primarily measured model collapse through generation quality: perplexity, output diversity, and qualitative inspection of generated text. Our work shifts the evaluation target from generation to *comprehension*—the model's ability to process inputs it has never seen, measured through log-probability assignment to held-out novel linguistic constructions. This distinction proves consequential: we find that comprehension and generation degrade on different timescales.

Linguistic novelty and distributional change. Novel linguistic constructions—slang, neologisms, code-switching—emerge and spread through social networks on timescales of months to years [Eisenstein et al., 2014, Stewart and Eisenstein, 2018]. Standard language model training corpora capture a snapshot of language at collection time; constructions that emerge after this cutoff are effectively out-of-distribution. Domain-specific language models [Nguyen et al., 2020] partially address temporal mismatch but do not resolve the broader concern that recursive training on model outputs may prevent future models from acquiring novel constructions at all.

3 Experimental Setup

3.1 Model and Training

We fine-tune GPT-2 (124M parameters; Radford et al. 2019) from HuggingFace pretrained weights on a 2M-token subset of WikiText-103 [Merity et al., 2017]. Fine-tuning uses AdamW [Loshchilov and Hutter, 2019] with cosine learning rate schedule, effective batch size 32 (micro-batch 4 with 8-step gradient accumulation), and context length 256 tokens. We conduct two training regimes:

- **Shallow:** 500 steps at learning rate 5×10^{-5}, simulating light fine-tuning where pretrained representations are largely preserved.
- **Deep:** 5,000 steps at learning rate 5×10^{-5} (primary) or 1×10^{-4} (aggressive), simulating thorough fine-tuning that may overcome pretrained representations.

All conditions are replicated with 3 random seeds (42, 137, 256); results are reported as mean $\pm$ standard deviation. All models start from the same pretrained GPT-2 weights at each generation to isolate the effect of data contamination from weight drift.

3.2 Data Splits

The WikiText-103 corpus is split into training (90%), validation (5%), and test (5%) sets. Validation data is used exclusively for checkpoint selection during training; all reported metrics are computed on the held-out test set to prevent information leakage.

3.3 Generational Loop

For each contamination ratio $r \in \{0\%, 10\%, 25\%, 50\%, 100\%\}$:

- **Generation 0:** Fine-tune GPT-2 on clean WikiText-103.
- **Generation n ($n \geq 1$):** Generate synthetic text from generation $n-1$'s model using nucleus sampling [Holtzman et al., 2020] (top-$p = 0.95$, $T = 1.0$). Mix with the base corpus at ratio r. Fine-tune a fresh GPT-2 on the mixed data.

We run 3 generations per arm. The 0% control uses different effective seeds per generation (seed $+1000\times$ generation) so that the control arm produces genuinely different models with real variance from random batch ordering, rather than deterministic replicas.

3.4 Evaluation Metrics

Log-probability gap (primary metric). For a model M, we compute mean per-token log-probability on two held-out sets: (i) 200 random passages of 128 tokens from the WikiText test split (*standard*), sampled with a fixed seed for reproducibility, and (ii) 51 hand-curated sentences containing novel linguistic constructions with correct completions filled in (*novel*). The gap is:

$$\Delta_{\text{LP}} = \bar{\ell}_{\text{standard}} - \bar{\ell}_{\text{novel}} \tag{1}$$

where $\bar{\ell}$ denotes mean per-token log-probability. Since novel constructions are harder, $\Delta_{\text{LP}} > 0$. A widening gap under contamination indicates that the model is becoming relatively worse at novel text compared to standard text.

Critical-span log-probability gap. A refined variant that computes $\bar{\ell}_{\text{novel}}$ only over the annotated critical span of each benchmark sentence—the tokens containing the novel construction itself—excluding surrounding context that may be conventional. This variant amplifies the contamination signal by focusing on the tokens that are genuinely out-of-distribution.

Test perplexity. Standard cross-entropy loss on the held-out WikiText test set, reported as $\exp(\text{loss})$.

Distinct-N. Unigram, bigram, and trigram type-to-token ratio over 50 prompted generations of 100 tokens each [Li et al., 2016].

Novelty accuracy. Multiple-choice cloze accuracy on the 51-example linguistic novelty benchmark. Each example presents a sentence with a blank and four options; the model selects the option with the highest mean log-probability over the completion span.

3.5 Linguistic Novelty Benchmark

The benchmark comprises 51 examples across five categories of linguistic phenomena that post-date or are absent from GPT-2's training distribution: novel slang (20 examples), creative neologisms (14), rhetorical subversion (8), pragmatic inference (6), and code-switching (3). Each example includes character-level annotations of the critical span containing the novel construction, enabling the critical-span gap variant.

3.6 Compute

All experiments are run on a single Apple M4 Pro (24 GB unified memory) using PyTorch's MPS backend. Table 1 summarizes the compute budget. Total wall-clock time across all experiments is approximately 75 hours, dominated by the deep fine-tuning multi-seed sweep.

Table 1: Compute budget. All runs on Apple M4 Pro, 24 GB RAM, MPS backend.

Phase	Runs	Time per run	Total
Shallow multi-seed	54	$\sim$25 min	14.5 h
Deep multi-seed	45 (30 trained)	$\sim$2.5 h	60.2 h
Critical-span re-eval	45	$<$1 min	$\sim$0.5 h
Total			**$\sim$75 h**

4 Results: Shallow Fine-Tuning

4.1 Log-Probability Gap Widens with Contamination

Table 2 presents the log-probability gap across all shallow fine-tuning conditions. The 0% control arm produces a stable gap of 0.49 ± 0.01 across all generations. Contaminated arms show statistically significant dose-dependent gap widening at generation 1, with the 100% arm producing a gap of 1.36 ± 0.10—nearly 3$\times$ the baseline. Figure 1 visualizes this trend with error bands across seeds.

Table 2: Log-probability gap (Δ_{LP}, mean $\pm$ std, n=3) under shallow fine-tuning (500 steps). Higher values indicate a wider differential between standard and novel text processing.

Contamination	Gen 0	Gen 1	Gen 2
0% (control)	0.490 ± 0.002	0.504 ± 0.017	0.492 ± 0.012
10%	0.490 ± 0.002	0.604 ± 0.012	0.589 ± 0.002
25%	0.490 ± 0.002	0.643 ± 0.014	0.614 ± 0.003
50%	0.490 ± 0.002	0.745 ± 0.011	0.660 ± 0.008
100%	0.490 ± 0.002	$\mathbf{1.363 \pm 0.103}$	$\mathbf{1.040 \pm 0.130}$
Compound (25%)	0.490 ± 0.002	0.643 ± 0.014	0.615 ± 0.001

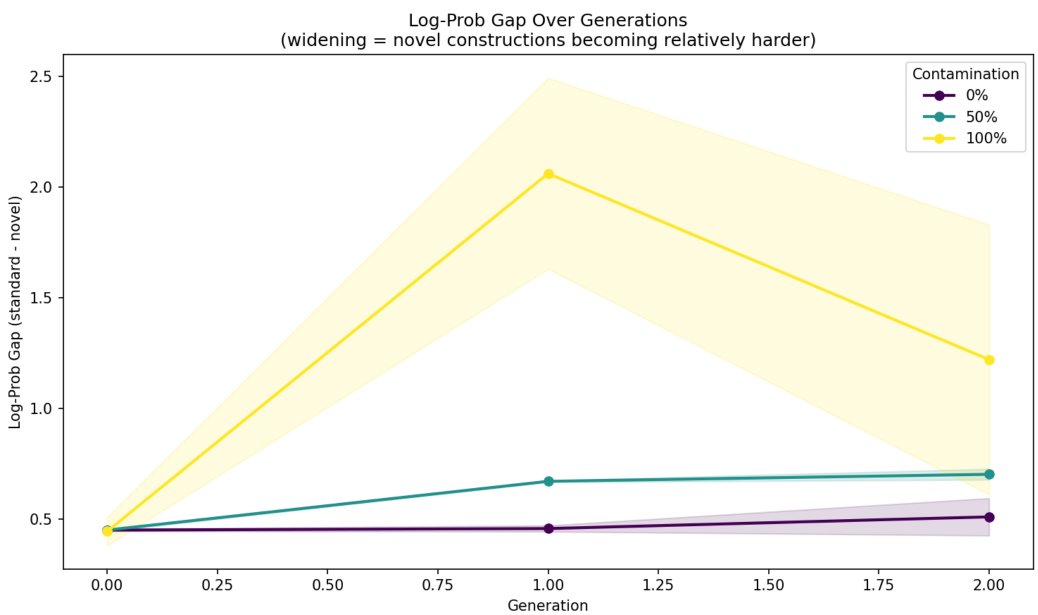

Figure 1: Log-probability gap across generations and contamination ratios under shallow fine-tuning. Error bands show $\pm$1 standard deviation over n=3 seeds. The dose-dependent widening is monotonic at generation 1; partial recovery at generation 2 reflects the degenerate collapse artifact at 100% contamination (Section 5.6).

4.2 Decomposing the Gap: Distribution Narrowing

Table 3 decomposes the gap into its standard and novel components at generation 1. At 10–50% contamination, the gap widens primarily because standard-text log-probability *improves* (from -5.72 toward -5.46) while novel-text log-probability remains approximately constant (~ -6.21). The model becomes more confident on in-distribution text without losing capacity for novel constructions—distribution *narrowing* rather than competence loss.

At 100% contamination, both mechanisms contribute: standard-text log-probability improves sharply ($-5.72 \rightarrow -5.08$) *and* novel-text log-probability degrades ($-6.21 \rightarrow -6.44$). This is the only condition under shallow training where novel-text processing genuinely worsens.

Table 3: Decomposition of mean log-probability at generation 1 (shallow, mean $\pm$ std, n=3). At moderate contamination, the gap widens because standard text gets easier; at 100%, novel text also gets harder.

Contamination	$\bar{\ell}_{\text{std}}$ (Gen 0)	$\bar{\ell}_{\text{std}}$ (Gen 1)	$\bar{\ell}_{\text{nov}}$ (Gen 0)	$\bar{\ell}_{\text{nov}}$ (Gen 1)
0%	-5.72 ± 0.01	-5.71 ± 0.03	-6.21 ± 0.01	-6.21 ± 0.03
10%	-5.72 ± 0.01	-5.62 ± 0.01	-6.21 ± 0.01	-6.22 ± 0.02
25%	-5.72 ± 0.01	-5.57 ± 0.01	-6.21 ± 0.01	-6.21 ± 0.02
50%	-5.72 ± 0.01	-5.46 ± 0.01	-6.21 ± 0.01	-6.20 ± 0.02
100%	-5.72 ± 0.01	$\mathbf{-5.08 \pm 0.14}$	-6.21 ± 0.01	$\mathbf{-6.44 \pm 0.25}$

4.3 Validation Perplexity

Perplexity degrades monotonically with contamination (Table 4), replicating the model collapse findings of Shumailov et al. [2024b]. At 10–50% contamination, the degradation is modest (5–22%). The 100% arm exhibits catastrophic collapse, reaching a perplexity of $1{,}697 \pm 112$ by generation 2.

Table 4: Validation perplexity (mean $\pm$ std, n=3) under shallow fine-tuning. The 100% arm shows catastrophic collapse; moderate contamination produces modest degradation.

Contamination	Gen 0	Gen 1	Gen 2
0%	247 ± 2	248 ± 4	248 ± 4
10%	247 ± 2	261 ± 2	261 ± 2
25%	247 ± 2	274 ± 2	273 ± 2
50%	247 ± 2	303 ± 2	300 ± 3
100%	247 ± 2	$\mathbf{954 \pm 12}$	$\mathbf{1{,}697 \pm 112}$

4.4 Generation Diversity Collapses

Distinct-1 (unigram diversity) provides the clearest experimental signal (Table 5 and Figure 2). Even 10% contamination reduces diversity from 0.43 to 0.27—a 37% drop—while the log-probability gap at the same condition shows only a modest increase from 0.49 to 0.60. This asymmetry between generation and comprehension degradation is a central finding: synthetic contamination first destroys generative diversity before affecting comprehension. At 100% contamination by generation 2, Distinct-1 falls to 0.04, indicating near-repetitive output. The 0% control remains stable across all generations.

4.5 Novelty Accuracy and Compounding

Multiple-choice novelty accuracy remained flat across all conditions (0.29–0.37), confirming that GPT-2 answers these forced-choice questions from token-frequency priors rather than pragmatic comprehension. This metric is too coarse to detect the effects captured by the continuous log-probability gap.

The compounding arm (25% contamination with model-weight inheritance across generations) produced results nearly identical to the fixed 25% arm: a generation-1 gap of 0.643 versus 0.643,

Table 5: Distinct-1 (mean $\pm$ std, n=3) under shallow fine-tuning. Diversity collapses at all contamination levels, providing the earliest signal of distribution narrowing.

Contamination	Gen 0	Gen 1	Gen 2
0%	0.426 ± 0.006	0.438 ± 0.006	0.441 ± 0.008
10%	0.428 ± 0.006	0.272 ± 0.008	0.267 ± 0.013
25%	0.428 ± 0.003	0.226 ± 0.027	0.182 ± 0.014
50%	0.429 ± 0.003	0.203 ± 0.010	0.164 ± 0.004
100%	0.431 ± 0.007	0.103 ± 0.004	$\mathbf{0.042 \pm 0.004}$

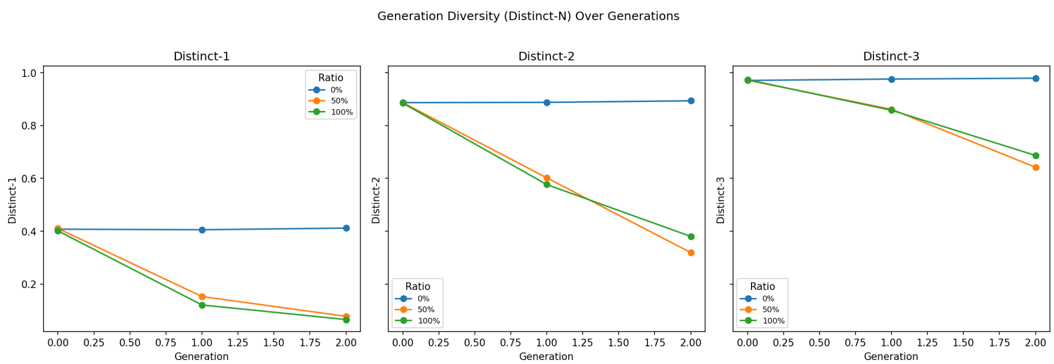

Figure 2: Distinct-N diversity metrics across generations and contamination ratios. Diversity collapses sharply even at 10% contamination, making it the most sensitive early-warning signal of distribution narrowing.

and a generation-2 gap of 0.615 versus 0.614. Over 3 generations, compounding confers no additional degradation beyond what fixed-ratio contamination produces, suggesting rapid saturation to a contaminated equilibrium.

5 Results: Deep Fine-Tuning Breaks the Pretrained Buffer

The shallow fine-tuning results raised a critical question: is the preservation of novel-text comprehension at moderate contamination a genuine property of the data, or merely an artifact of insufficient optimization? GPT-2's pretrained representations may act as a buffer that resists contamination under 500 steps of fine-tuning but would yield under more aggressive training. We test this hypothesis by extending fine-tuning to 5,000 steps (10$\times$ longer) at contamination ratios of 0%, 50%, and 100%.

5.1 The Gap Under Deep Training

Table 6 presents the deep fine-tuning results at learning rate 5×10^{-5}. At 100% contamination, generation 1, the gap widens to $\mathbf{2.30} \pm \mathbf{0.09}$ ($p < 0.001$ vs. control), compared to 1.36 ± 0.10 under shallow training. Deeper optimization drives the model further from its pretrained representations, producing a $1.7\times$ larger gap.

At 50% contamination, the gap also widens significantly (0.67 ± 0.004, $p < 0.001$), and unlike the shallow regime, the gap continues to grow at generation 2 (0.70 ± 0.02, $p = 0.036$) rather than recovering.

5.2 Perplexity Confirms Monotonic Collapse

While the log-probability gap shows an apparent "recovery" at 100% generation 2, perplexity tells the true story (Table 7): monotonic degradation from $192 \rightarrow 854 \rightarrow 1{,}482$. The gap compression at generation 2 is a metric artifact, not model improvement (Section 5.6). Figure 3 visualizes the relationship between perplexity and gap across all conditions, revealing the degenerate region where the two metrics diverge.

Table 6: Log-probability gap (Δ_{LP}, mean $\pm$ std) under deep fine-tuning (5,000 steps, LR $= 5 \times 10^{-5}$, n=3). Asterisks denote significance vs. 0% control ($^*p < 0.05$, $^{**}p < 0.001$).

Contamination	Gen 0	Gen 1	Gen 2
0% (control)	0.449 ± 0.005	0.456 ± 0.014	0.509 ± 0.084
50%	0.449 ± 0.005	$0.669 \pm 0.004^{**}$	$0.701 \pm 0.024^{*}$
100%	0.449 ± 0.005	$\mathbf{2.297 \pm 0.091}^{**}$	$0.917 \pm 0.021^{\dagger}$

†Not significant ($p = 0.12$). See Section 5.6 for discussion of the degenerate collapse artifact.

Table 7: Validation perplexity (mean $\pm$ std, n=3) under deep fine-tuning. Monotonic collapse at 100% contamination confirms progressive model degradation despite apparent gap "recovery."

Contamination	Gen 0	Gen 1	Gen 2
0%	192 ± 0	192 ± 0	192 ± 1
50%	192 ± 0	233 ± 1	236 ± 0
100%	192 ± 0	$\mathbf{854 \pm 23}$	$\mathbf{1{,}482 \pm 30}$

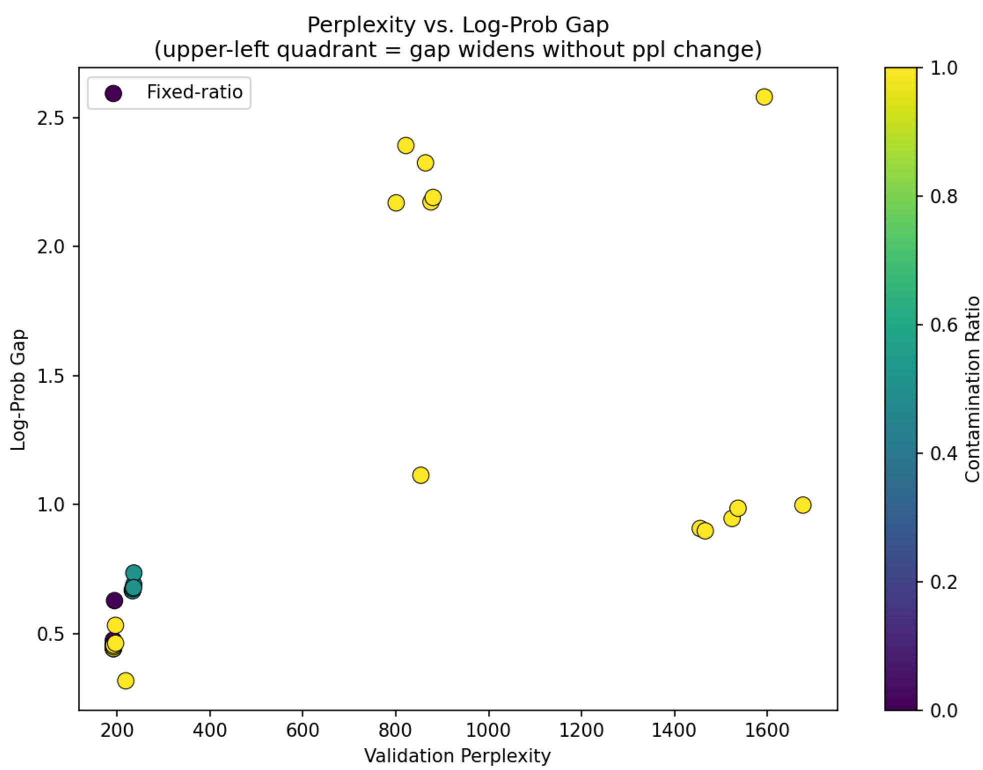

Figure 3: Perplexity vs. log-probability gap across all conditions. Below PPL $\approx 1{,}000$, the gap correlates positively with perplexity. Above PPL $\approx 1{,}400$, the model enters degenerate collapse and the gap compresses despite worsening perplexity.

5.3 Aggressive Learning Rate Amplifies Degradation

To further test whether deeper optimization breaks the pretrained buffer, we repeated the 100% arm with a $2\times$ higher learning rate (1×10^{-4}). Table 8 shows qualitatively similar results: the gap rises to 1.82 ± 0.50 at generation 1 and 1.52 ± 0.75 at generation 2, with perplexity reaching $1{,}603 \pm 57$. The higher variance reflects the instability of aggressive optimization on contaminated data. Figure 4 compares the two learning rate configurations directly.

Table 8: Aggressive learning rate (1×10^{-4}) at 100% contamination (mean $\pm$ std, n=3). Higher LR produces qualitatively similar degradation with greater variance.

Gen	Δ_{LP}	$\Delta_{\text{LP}}^{\text{crit}}$	PPL	D-1
0	0.437 ± 0.090	1.147 ± 0.105	204 ± 10	0.391 ± 0.006
1	$\mathbf{1.825 \pm 0.503}$	$\mathbf{2.797 \pm 0.704}$	$\mathbf{845 \pm 33}$	0.122 ± 0.017
2	1.521 ± 0.749	2.283 ± 0.917	$\mathbf{1{,}603 \pm 57}$	0.054 ± 0.008

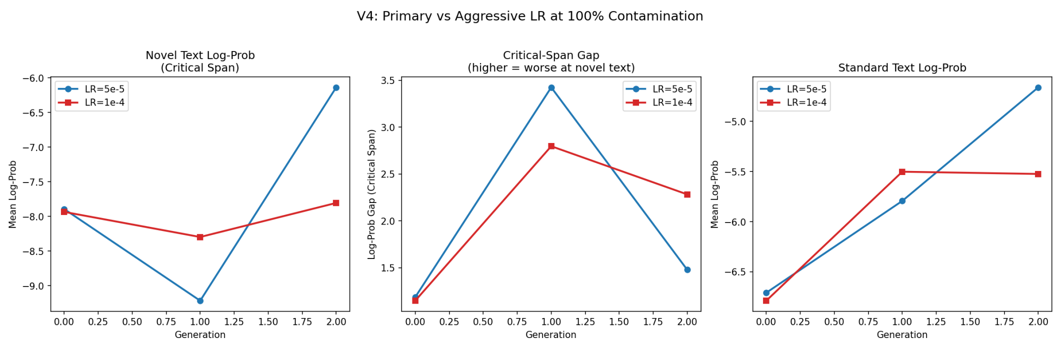

Figure 4: Comparison of primary (LR $= 5 \times 10^{-5}$) and aggressive (LR $= 1 \times 10^{-4}$) configurations at 100% contamination under deep fine-tuning. Both configurations produce progressive degradation; the aggressive LR shows higher variance.

5.4 Decomposition: Deep Training Degrades Novel-Text Processing

The critical difference between shallow and deep fine-tuning is visible in the decomposition (Table 9 and Figure 5). Under deep training at 100% contamination generation 1, novel-text log-probability drops to -8.09 ± 0.06, compared to -6.44 ± 0.25 under shallow training—a 1.65 nats further decline. The pretrained buffer that preserved novel-text processing under shallow training has been overcome: 5,000 steps of optimization on contaminated data is sufficient to degrade the model's internal representations of out-of-distribution language.

Table 9: Decomposition at generation 1 under deep fine-tuning (primary LR, mean $\pm$ std, n=3). Novel-text log-probability degrades sharply at 100%, unlike under shallow training where it remained stable at moderate contamination.

Contamination	$\bar{\ell}_{\text{std}}$ (Gen 0)	$\bar{\ell}_{\text{std}}$ (Gen 1)	$\bar{\ell}_{\text{nov}}$ (Gen 0)	$\bar{\ell}_{\text{nov}}$ (Gen 1)
0%	-6.71 ± 0.02	-6.71 ± 0.00	-7.16 ± 0.02	-7.17 ± 0.01
50%	-6.71 ± 0.02	-6.42 ± 0.02	-7.16 ± 0.02	-7.09 ± 0.03
100%	-6.71 ± 0.02	$\mathbf{-5.79 \pm 0.10}$	-7.16 ± 0.02	$\mathbf{-8.09 \pm 0.06}$

5.5 Critical-Span Gap Amplifies the Signal

The critical-span variant of the gap, which scores only the novel construction tokens rather than the full sentence, consistently amplifies the contamination signal (Figures 6 and 7). Under deep training at 100% contamination generation 1, the critical-span gap reaches 3.43 ± 0.11 compared to 2.30 ± 0.09

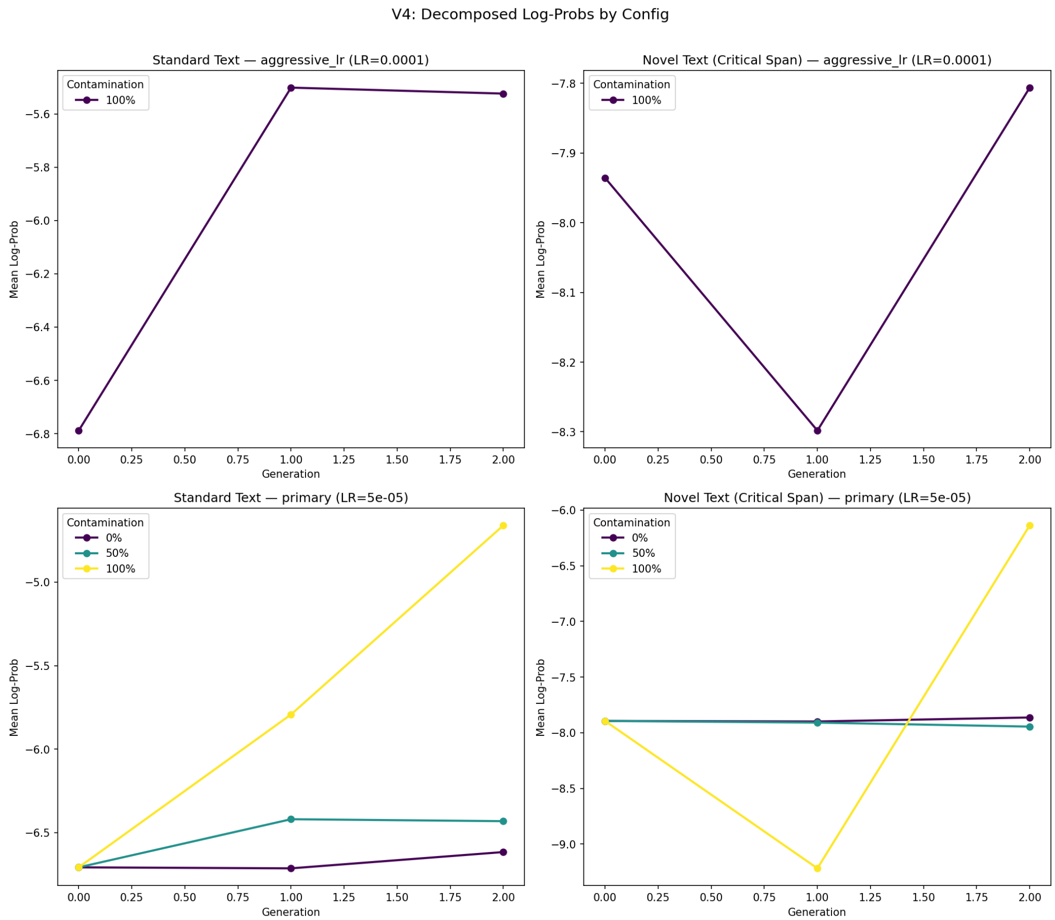

Figure 5: Decomposed log-probability components under deep fine-tuning. Standard-text log-probability (dashed) improves at all contamination levels; novel-text log-probability (solid) degrades sharply at 100%, revealing that the pretrained buffer has been overcome.

for the full-sentence gap. This confirms that the novel construction tokens themselves—not the surrounding conventional context—are the locus of degradation.

5.6 The Degenerate Collapse Artifact

At 100% contamination generation 2 under deep training, the primary configuration shows an apparent "recovery" in the log-probability gap (0.92 ± 0.02), despite perplexity reaching $1{,}482 \pm 30$. This is a metric artifact, not model improvement.

The mechanism is as follows. By generation 2, the model's probability distribution has collapsed to a narrow spike over a small set of memorized patterns. Standard-text log-probability jumps from its baseline of -6.71 to -4.66: the model assigns absurdly high confidence to the few patterns it has memorized. Novel-text log-probability is also pulled upward (from -7.16 to -5.58) as both scores compress toward the distorted distribution. The gap shrinks because both components are pulled toward the same degenerate attractor, not because the model has recovered any competence.

We identify perplexity $> 1{,}400$ as a practical threshold for degenerate collapse, beyond which log-probability gap metrics should not be interpreted at face value. Perplexity itself remains a reliable monotonic indicator of degradation throughout.

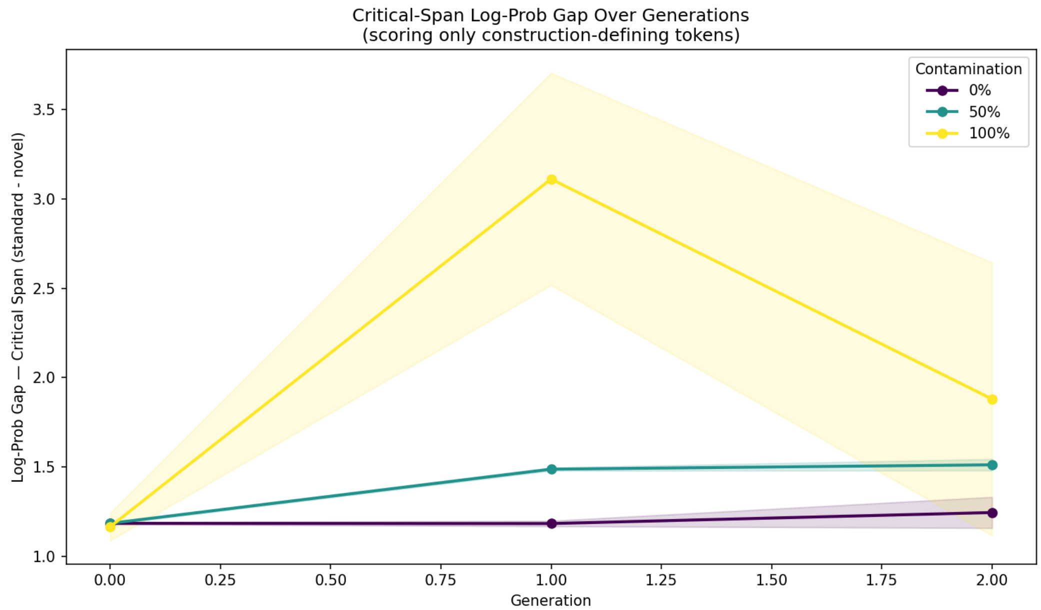

Figure 6: Critical-span log-probability gap, measuring only the tokens within annotated novel construction spans. The contamination signal is amplified relative to the full-sentence gap (Figure 1).

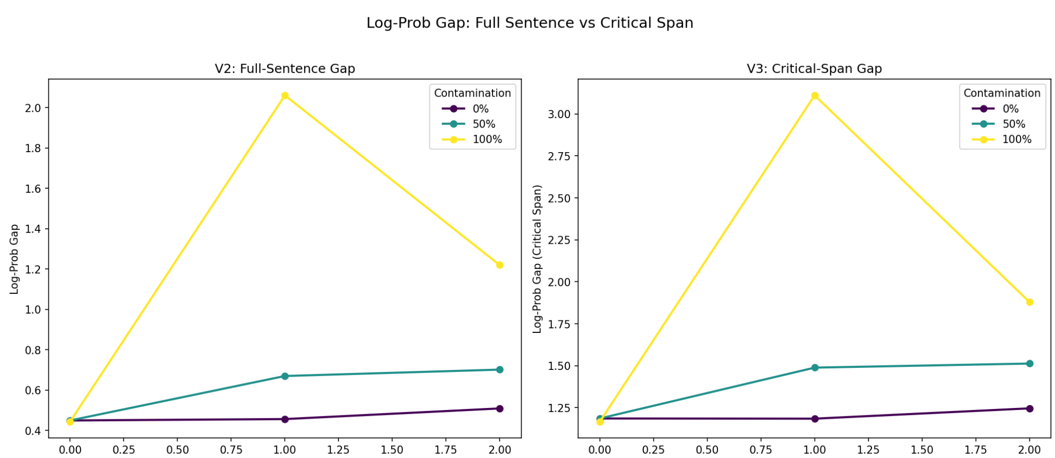

Figure 7: Comparison of full-sentence and critical-span gap variants. The critical-span gap (right) is uniformly larger and shows clearer separation between contamination levels, confirming that novel construction tokens are the primary locus of degradation.

6 Discussion

6.1 Training Dynamics

Figure 8 shows validation loss curves for generation 1 models at different contamination levels. Under shallow training, all conditions converge within 500 steps, with contaminated arms settling to higher loss values. Under deep training, the 0% and 50% arms converge smoothly, but the 100% arm exhibits persistently elevated loss throughout 5,000 steps—the model never recovers from the distribution mismatch introduced by fully synthetic training data. The aggressive learning rate configuration at 100% shows even higher loss, confirming that stronger optimization on contaminated data worsens rather than overcomes the distribution shift.

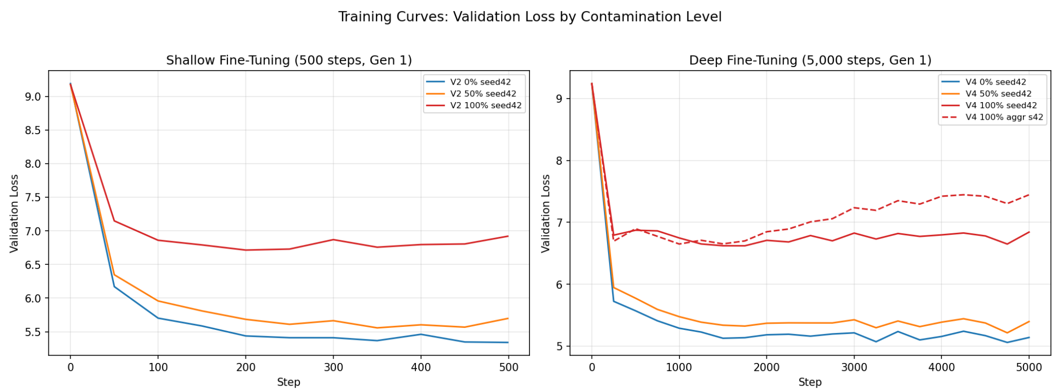

Figure 8: Validation loss curves for generation 1 models under shallow (left) and deep (right) fine-tuning. Under deep training, the 100% contamination arms settle to substantially higher loss, reflecting persistent distribution mismatch from fully synthetic data.

6.2 Two Regimes of Degradation

Our results reveal two distinct contamination regimes, mediated by optimization depth:

1. **Shallow training + moderate contamination:** The model develops a narrower but more confident distribution over standard text. Novel constructions are not degraded; they are left behind as the model specializes. This is distribution *narrowing*—the model becomes a specialist rather than a generalist—but not competence loss.
2. **Deep training + high contamination:** The pretrained buffer breaks. Novel-text log-probabilities degrade sharply ($-7.16 \rightarrow -8.09$), and perplexity collapses into the degenerate regime. This is genuine competence *loss*: the model's internal representations of out-of-distribution language are overwritten by optimization on contaminated data. This is consistent with the finding of Kumar et al. [2022] that standard fine-tuning can distort pretrained features, and with the catastrophic forgetting literature [Kirkpatrick et al., 2017] more broadly.

The practical implication is that pretrained representations offer meaningful but *not unlimited* protection against synthetic contamination. Light fine-tuning on moderately contaminated data may be safe; thorough fine-tuning on heavily contaminated data is not. This motivates the development of contamination detection methods [Shi et al., 2024, Golchin and Surdeanu, 2024] as a preprocessing step before fine-tuning on web-scraped data.

6.3 Diversity Collapse as an Early Warning

Generation diversity proved to be the most sensitive indicator of distribution narrowing across both training regimes, collapsing even at 10% contamination under shallow training while the log-probability gap at the same condition showed only modest movement. Distinct-N monitoring could serve as a practical early-warning signal for synthetic contamination in training pipelines: a sharp drop relative to the pretrained baseline may indicate distributional narrowing before it manifests in comprehension metrics or downstream task performance.

6.4 Comprehension vs. Generation: An Asymmetry

A central finding is the asymmetry between generation and comprehension degradation. Under shallow training at 10% contamination, Distinct-1 drops 37% while the log-probability gap increases by only 23%. This suggests that synthetic contamination first destroys generative diversity—the model's ability to *produce* varied output—before affecting comprehension—its ability to *process* varied input. The asymmetry narrows under deep training, where both generation and comprehension degrade together, consistent with the pretrained buffer breaking down.

6.5 Limitations

1. **Small model.** GPT-2 124M may not be representative of larger models, which have deeper and more robust pretrained representations [Brown et al., 2020, Touvron et al., 2023]. The buffer may be substantially more durable in models with billions of parameters, consistent with scaling law predictions that larger models learn more robust features [Kaplan et al., 2020, Hoffmann et al., 2022].
2. **Only 3 generations.** Shumailov et al. [2024b] used up to 10 generations. Additional generations may reveal whether moderate contamination eventually breaks the buffer even under shallow training.
3. **Self-contamination only.** All synthetic data is generated by variants of the same architecture. Cross-model contamination (e.g., training GPT-2 on data generated by a different model family) might produce different dynamics.
4. **Benchmark size.** With 51 examples across 5 categories, the benchmark provides limited statistical power for per-category analysis. Categories with 3–6 items (code-switching, pragmatic inference) cannot support reliable subcategory conclusions. We report a preliminary per-category breakdown in Section A but caution against strong claims from small-n categories.
5. **Token-level metric.** The log-probability gap measures token prediction rather than pragmatic comprehension directly. A model that assigns low probability to novel tokens may still "comprehend" the construction in some functional sense not captured by this metric.

7 Conclusion

We tested whether synthetic data contamination destroys a language model's ability to process novel linguistic constructions, using multi-seed replication (n=3) across two training depths.

Under shallow fine-tuning, moderate contamination (10–50%) produces distribution *narrowing*: the model specializes toward conventional text without losing capacity for novel constructions. Under deep fine-tuning, the pretrained buffer breaks: at 100% contamination, the log-probability gap widens to 2.30 ± 0.09 ($p < 0.001$) with perplexity collapsing from 192 to 1,482 across two generations of recursive training.

These findings establish that pretrained representations are a buffer, not a firewall. They protect novel-text comprehension under light training but yield under sufficient optimization pressure on contaminated data. Generation diversity emerges as the earliest and most sensitive indicator of distributional narrowing, collapsing at contamination levels where comprehension metrics show no effect.

We recommend Distinct-N monitoring as a practical diagnostic for synthetic contamination in training pipelines, and we caution against interpreting log-probability gap metrics when model perplexity exceeds the degenerate collapse threshold (PPL $>$ 1,400). Future work should extend this analysis to larger models with deeper pretrained buffers, more generations of recursive training, and a finer contamination sweep in the 50–100% range to precisely locate the threshold at which narrowing transitions to competence loss.

Broader Impact

This work studies how synthetic data contamination affects language model capabilities, with implications for the trajectory of model training as machine-generated text becomes an increasing fraction of the web.

Risks to linguistic diversity. Our central finding—that contaminated models narrow toward conventional language patterns—has direct implications for speakers of non-standard language varieties. If future models are trained on data contaminated by prior model outputs, the resulting distribution narrowing may systematically disadvantage novel slang, code-switching, creative neologisms, and other forms of linguistic innovation. Communities whose language practices are underrepresented in training corpora are most at risk, as their constructions are furthest from the distribution that contaminated models reinforce.

Training pipeline implications. As the cost of generating synthetic text falls, the temptation to augment training corpora with model outputs grows. Our results suggest that moderate contamination (10–50%) under light fine-tuning may be relatively safe for comprehension tasks, but practitioners should monitor generation diversity (Distinct-N) as an early warning signal. Heavy contamination under deep training degrades the pretrained representations themselves, and it is unclear whether subsequent clean fine-tuning could recover the lost capacity.

Scope limitations. This study uses a single small model (GPT-2 124M), English-only data, and a 51-item benchmark. We caution against extrapolating directly to production-scale models or multilingual settings. The degenerate collapse threshold we identify (PPL $> 1{,}400$) is specific to our experimental configuration and should not be treated as a universal constant.

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Appendix

A Per-Category Sensitivity Analysis

To investigate whether certain types of novel constructions are more vulnerable to contamination than others, we compute the critical-span gap change from baseline (0% contamination, generation 0) to the most degraded non-degenerate condition (100% contamination, generation 1) under deep fine-tuning. Table 10 reports the results per category, and Figures 9 and 10 visualize the patterns.

All five categories show substantial degradation, with critical-span gaps increasing by 160–260% relative to baseline. Rhetorical subversion shows the largest relative increase (+262%), suggesting that pragmatically complex constructions—where meaning depends on subverting the expected register or tone—are particularly fragile under contamination. Creative neologisms show the smallest relative increase (+158%), possibly because novel word forms are already poorly modeled at baseline, leaving less room for further degradation.

These results should be interpreted cautiously given the small per-category sample sizes, particularly for code-switching (n=3) and pragmatic inference (n=6).

Table 10: Per-category critical-span gap sensitivity under deep fine-tuning (primary LR, mean over n=3 seeds). Δ is the change from baseline (0%, Gen 0) to contaminated (100%, Gen 1).

Category	N	Gap (0%, G0)	Gap (100%, G1)	Δ	Rel.
Rhetorical subversion	8	0.904	3.270	+2.366	+262%
Code-switching	3	1.101	3.470	+2.368	+215%
Pragmatic inference	6	1.270	3.565	+2.296	+181%
Novel slang	20	1.333	3.586	+2.253	+169%
Creative neologism	14	1.242	3.203	+1.961	+158%

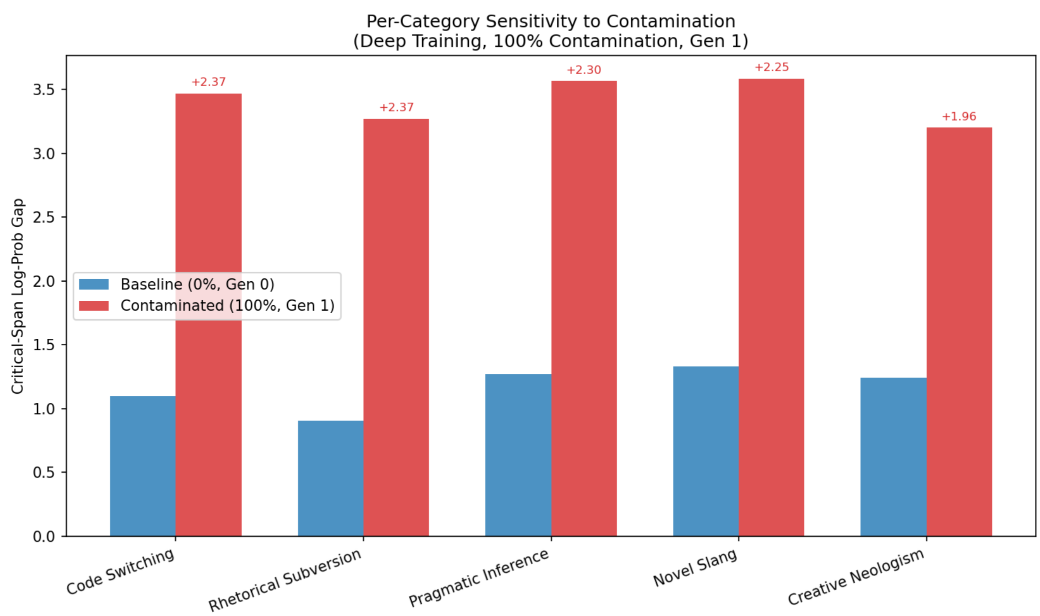

Figure 9: Per-category critical-span gap at baseline vs. 100% contamination generation 1 under deep fine-tuning. All categories show substantial degradation; annotations show the absolute gap increase.

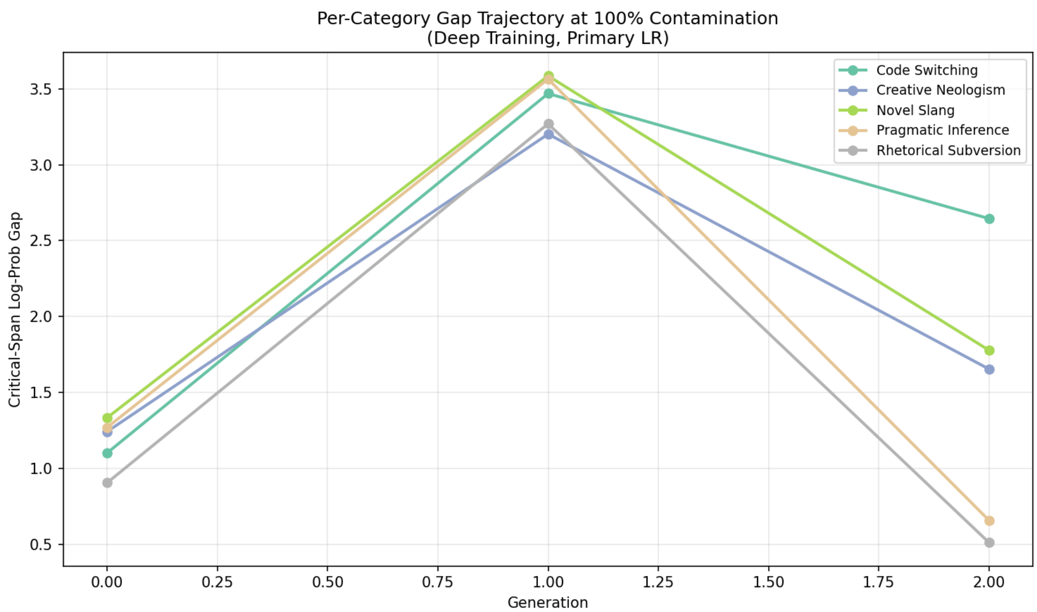

Figure 10: Per-category critical-span gap trajectory across generations at 100% contamination (deep fine-tuning, primary LR). All categories peak at generation 1 and compress at generation 2 due to the degenerate collapse artifact.